Lecture 15: Nonlinear Resistive Circuits

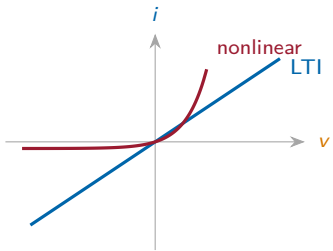
EE201 – Circuit Theory I

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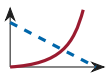
current voltage element characteristic - - - load line

Review: an element is a curve



- A resistor: a curve in the v–i plane
- LTI: a line through the origin
- Nonlinear: any other curve, e.g. $i = g(v)$
- So far: LTI circuits only

Roadmap



Load line

1

operating point

none or many



Substitution

2

Thévenin first

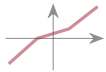


Ideal diode

3

on and off

series, parallel



Piecewise-linear circuits

4

i–*v* of a circuit

circuit from *i*–*v*

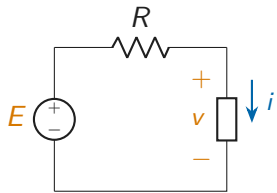


Time-varying

5

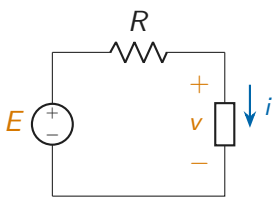
superposition

Motivation: nonlinear elements



- Diodes, lamps, transistors: nonlinear
- Node and mesh analysis: linear equations
- Now: one nonlinear equation, or a few linear pieces

One nonlinear resistor



Element: $i = g(v)$

KVL: $E = R\,i + v$

Algebraic solution

Solve $E = R\,g(v) + v$ for v_o

Then $i_o = g(v_o)$

Often g is only a measured curve

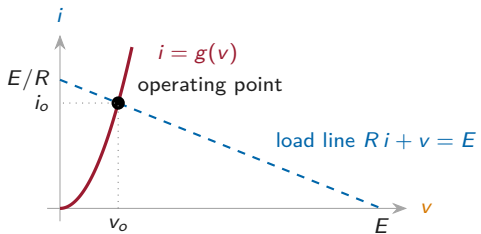
The load line

Two constraints on (v, i):

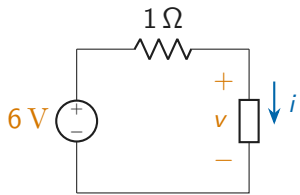
circuit: $R\,i + v = E$

element: $i = g(v)$

The solution is on both



Example 15.1: the operating point



$$i = \begin{cases} v^2, & v \geq 0 \\ 0, & v < 0 \end{cases} \quad \text{(A, V)}$$

$$(v_o, i_o) = ?$$

Example 15.1: two cases

KVL: $i + v - 6 = 0$

$v < 0$**:** $i = 0 \;\Rightarrow\; v = 6$

contradiction: no solution

Example 15.1: two cases

KVL: $i + v - 6 = 0$

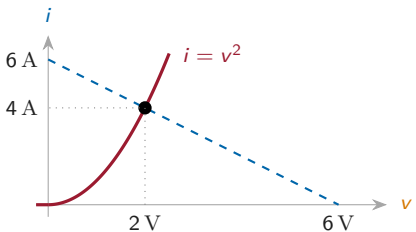
$v < 0$**:** $i = 0 \;\Rightarrow\; v = 6$

contradiction: no solution

$v \geq 0$**:** $v^2 + v - 6 = 0$

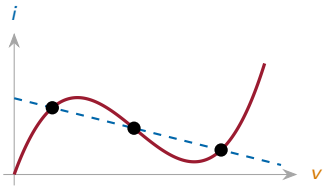
$(v-2)(v+3) = 0 \;\Rightarrow\; v = 2\,\mathrm{V}$

$$\boxed{(v_o, i_o) = (2\,\mathrm{V},\ 4\,\mathrm{A})}$$



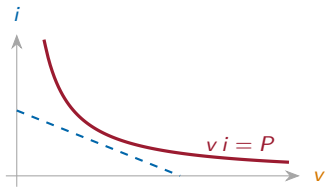
None, one or many operating points

Many



A bent curve: three intersections

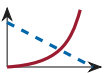
None



No intersection

A nonlinear circuit need not have exactly one solution

Roadmap



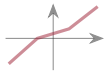
Load line



Substitution



Ideal diode



Piecewise-linear circuits



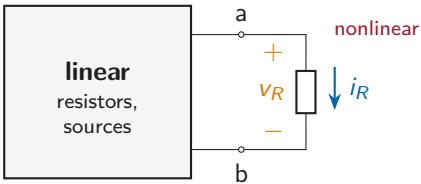
Time-varying

✓ 2 3 4 5

operating point ✓ | **Thévenin first** | on and off | i–v of a circuit | superposition

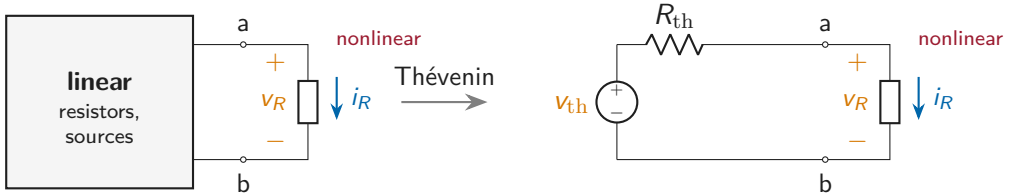
none or many ✓ | series, parallel | circuit from i–v

A nonlinear resistor in a larger circuit



One nonlinear resistor; the rest of the circuit is linear

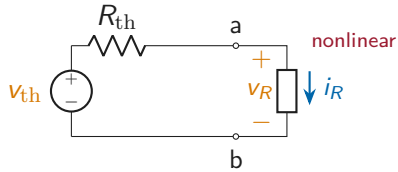
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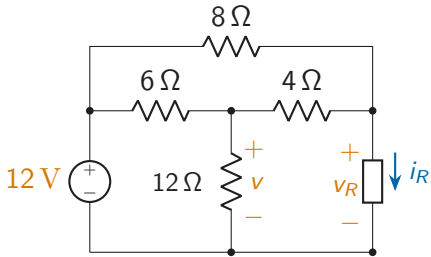
Seen from a–b, the rest is a source and a resistor: the case we just solved

Three steps



- **Step 1:** the Thévenin equivalent seen from a–b
- **Step 2:** the load line gives the operating point (v_R, i_R)
- **Step 3:** substitution theorem: replace the nonlinear resistor by a source v_R (or i_R)
- No other voltage or current changes
- Any other voltage or current then follows from a linear circuit

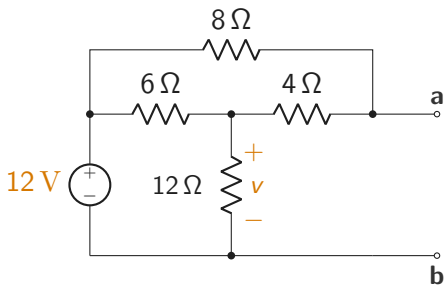
Example 15.2: find v



$$i_R = \begin{cases} \frac{1}{2} {v_R}^2, & v_R \geq 0 \\ 0, & v_R < 0 \end{cases}$$

$v = ?$

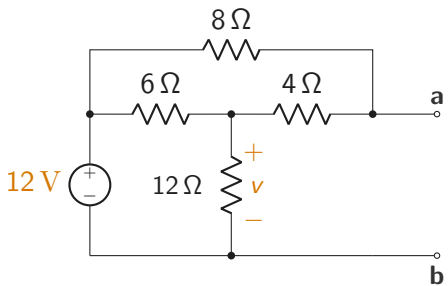
Example 15.2: step 1, Thévenin



Kill the source:

$R_{\text{th}} = 8 \parallel (4+6 \parallel 12) = 8 \parallel 8 = 4\,\Omega$

Example 15.2: step 1, Thévenin

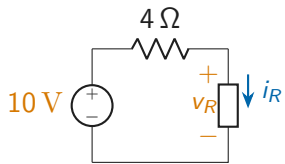


Kill the source:

$R_{\text{th}} = 8 \parallel (4+6 \parallel 12) = 8 \parallel 8 = 4\,\Omega$

Open circuit: two node equations

$e_1 = 9\,\text{V}, \quad v_{\text{th}} = v_{\text{oc}} = e_a = 10\,\text{V}$



Example 15.2: step 2, the operating point

KVL: $4\, i_R + v_R - 10 = 0$

Plug $i_R = \frac{1}{2}\, {v_R}^2$:

$2\, {v_R}^2 + v_R - 10 = 0$

$(2\, v_R + 5)(v_R - 2) = 0$

Example 15.2: step 2, the operating point

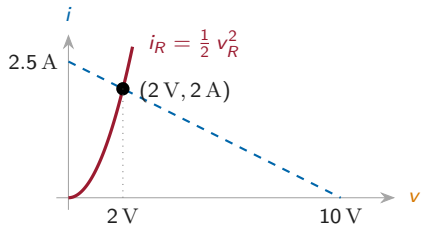
KVL: $4\,i_R + v_R - 10 = 0$

Plug $i_R = \frac{1}{2}\,{v_R}^2$:

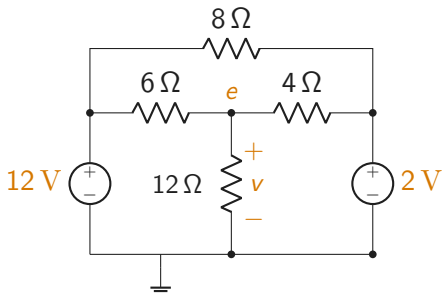
$2\,{v_R}^2 + v_R - 10 = 0$

$(2\,v_R + 5)(v_R - 2) = 0$

$v_R \geq 0$: $\quad v_R = 2\,\mathrm{V}, \quad i_R = 2\,\mathrm{A}$



Example 15.2: step 3, substitution



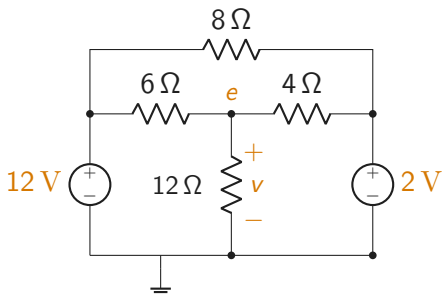
Replace it by a 2 V source

Node *e*:

$$\frac{e-12}{6}+\frac{e}{12}+\frac{e-2}{4}=0$$

$$\times 12:\quad 6e-30=0$$

Example 15.2: step 3, substitution



Replace it by a $2\,\mathrm{V}$ source

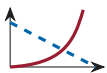
Node *e*:

$$\frac{e-12}{6}+\frac{e}{12}+\frac{e-2}{4}=0$$

$$\times 12:\quad 6\,e-30=0$$

$$\boxed{v=e=5\,\mathrm{V}}$$

Roadmap



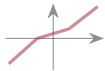
Load line



Substitution



Ideal diode



Piecewise-linear circuits



Time-varying

✓ ✓ 3 4 5

operating point ✓

none or many ✓

Thévenin first ✓

on and off

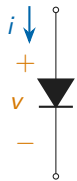
series, parallel

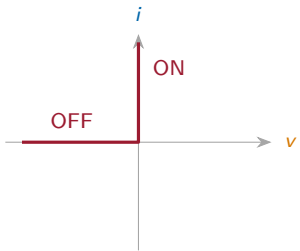
i–*v* of a circuit

circuit from *i*–*v*

superposition

The ideal diode





ON: $v = 0, \quad i \geq 0$

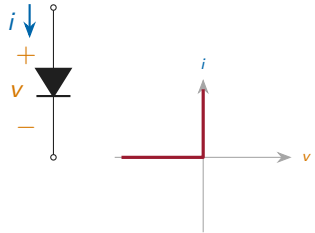
a short circuit

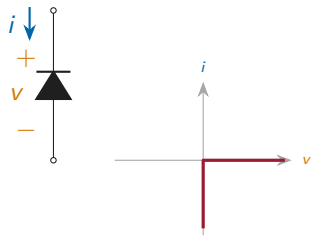
OFF: $i = 0, \quad v \leq 0$

an open circuit

Two straight pieces:
piecewise linear

Turning an element around





Swap the terminals: $v \to -v$, $i \to -i$

The curve is reflected through the origin

Series and parallel

Series: the same i

add the voltages

$v = v_a + v_b$

add the curves horizontally

Parallel: the same v

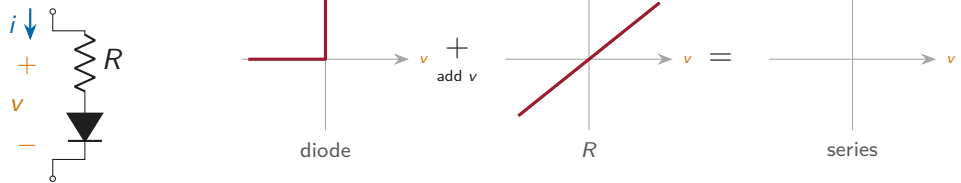
add the currents

$i = i_a + i_b$

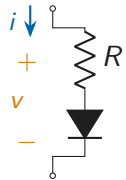
add the curves vertically

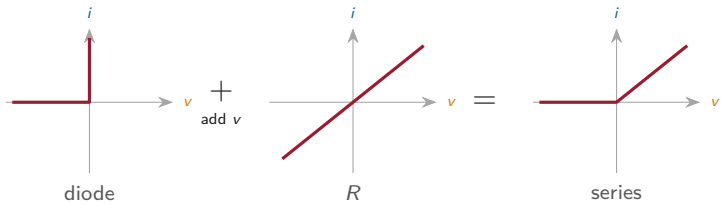
Diodes, LTI resistors and constant sources: the result is piecewise-linear

Case I: a resistor and a diode in series



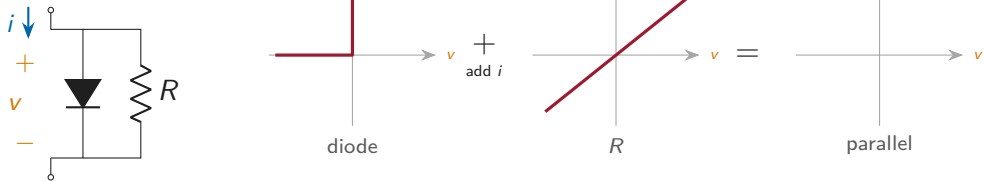
Case I: a resistor and a diode in series



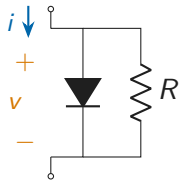


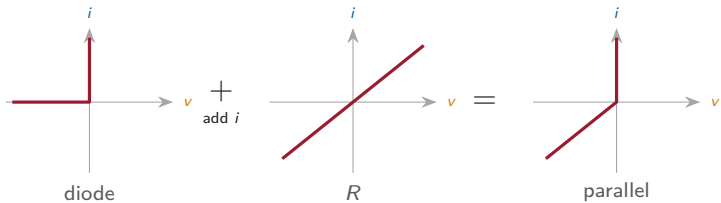
$$i > 0: \quad v = R\,i; \qquad i = 0: \quad v \leq 0$$

Case II: a resistor and a diode in parallel



Case II: a resistor and a diode in parallel

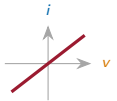




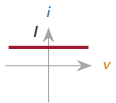
$v < 0$: $i = v/R$; $v = 0$: any $i \geq 0$

Reminder: voltage- or current-controlled

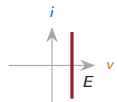
voltage-controlled	$i = g(v)$	vertical lines: once	$\rightarrow$ KCL, node analysis
current-controlled	$v = r(i)$	horizontal lines: once	$\rightarrow$ KVL, mesh analysis
neither			$\rightarrow$ assume ON or OFF



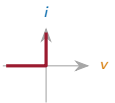
resistor



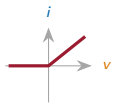
current source



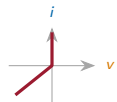
voltage source



ideal diode



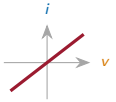
Case I: R in series



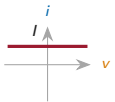
Case II: R in parallel

Reminder: voltage- or current-controlled

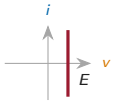
voltage-controlled	$i = g(v)$	vertical lines: once	$\rightarrow$ KCL, node analysis
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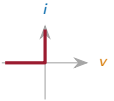
resistor
both



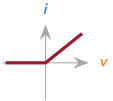
current source
voltage-controlled



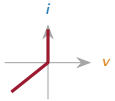
voltage source
current-controlled



ideal diode
neither

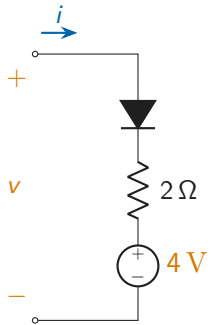


Case I: R in series
voltage-controlled



Case II: R in parallel
current-controlled

Example 15.3: a diode, a resistor and a source



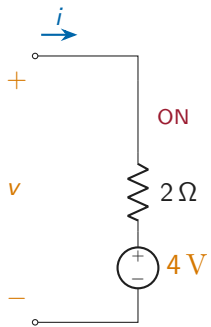
i as a function of v $= ?$

Method: assume a state for the diode

Solve the linear circuit

Keep the result where the assumption holds

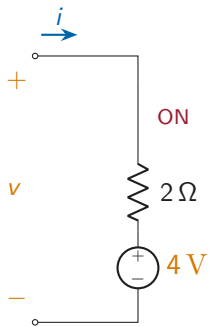
Example 15.3: assume the diode ON



ON: a short; condition $i \geq 0$

KVL: $v = 2\,i + 4$

Example 15.3: assume the diode ON

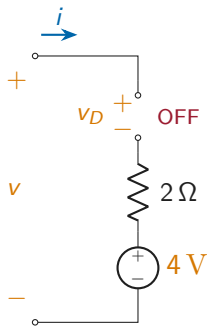


ON: a short; condition $i \geq 0$

KVL: $v = 2\,i + 4$

$$i = \frac{v - 4}{2} \geq 0 \;\Leftrightarrow\; v \geq 4\,\mathrm{V}$$

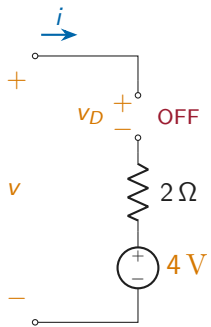
Example 15.3: assume the diode OFF



OFF: an open; condition $v_D < 0$

$i = 0$: no voltage on $2\,\Omega$

Example 15.3: assume the diode OFF



OFF: an open; condition $v_D < 0$

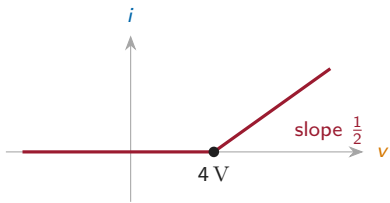
$i = 0$: no voltage on $2\,\Omega$

KVL: $v_D = v - 4$

$v_D < 0 \;\Leftrightarrow\; v < 4\,\mathrm{V}$

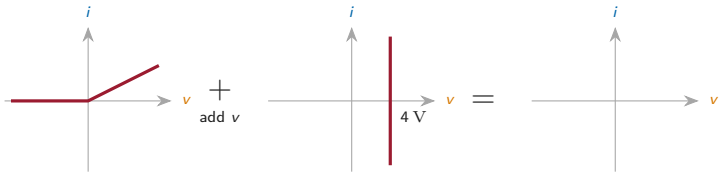
Example 15.3: the result

$$i = \begin{cases} \dfrac{v-4}{2}, & v \geq 4\,\mathrm{V} \\ 0, & v < 4\,\mathrm{V} \end{cases}$$

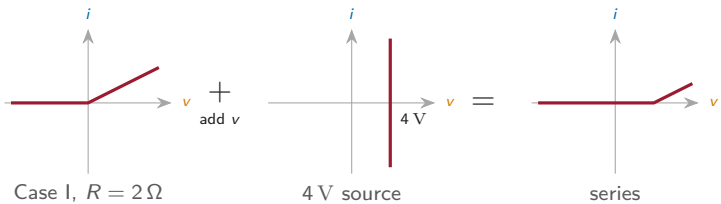


Breakpoint: $i_D = 0$ and $v_D = 0$

Example 15.3: graphically

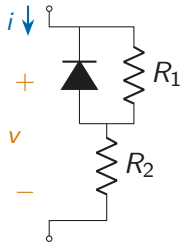


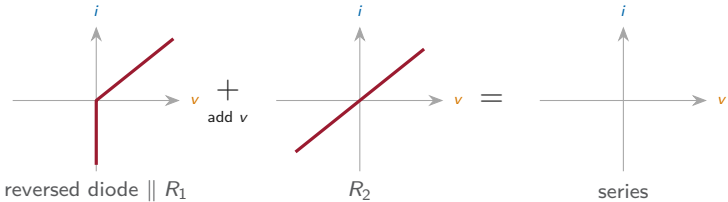
Example 15.3: graphically



A source in series shifts the curve right by $4\,\mathrm{V}$ ✓

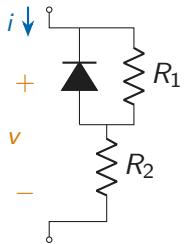
Case III: combinations

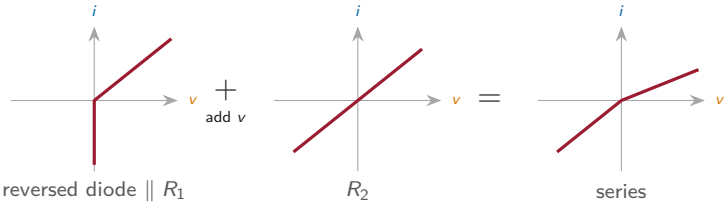




For simplicity, $R_1 = R_2 = R$.

Case III: combinations

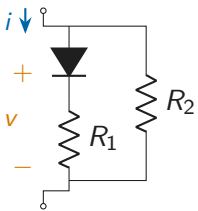


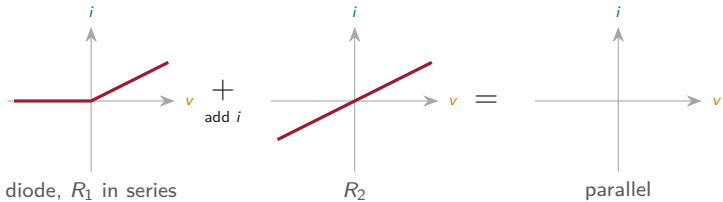


For simplicity, $R_1 = R_2 = R$.

$i > 0$: diode OFF, slope $\dfrac{1}{2R}$; $\quad i < 0$: diode ON, slope $\dfrac{1}{R}$

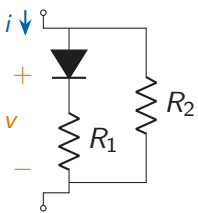
Case IV: combinations

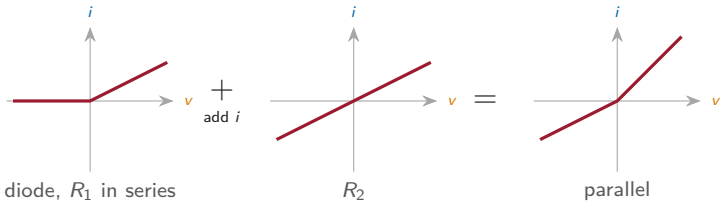




For simplicity, $R_1 = R_2 = R$.

Case IV: combinations

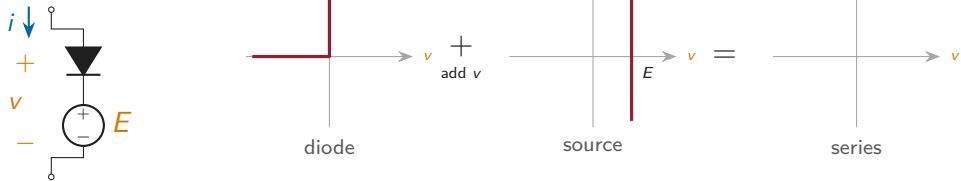




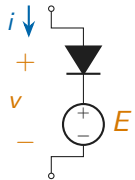
For simplicity, $R_1 = R_2 = R$.

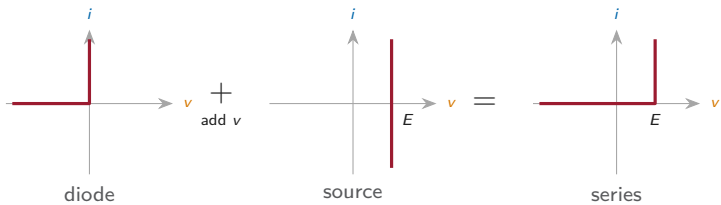
$v > 0$: slope $\frac{2}{R}$; $\quad v < 0$: slope $\frac{1}{R}$

Case V: a diode and a voltage source in series



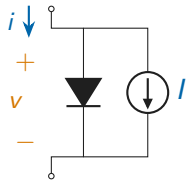
Case V: a diode and a voltage source in series

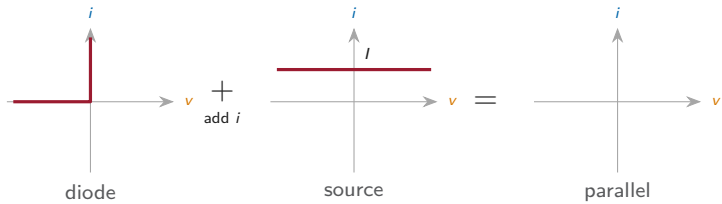




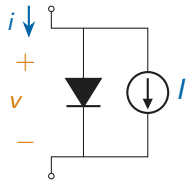
The diode curve shifted right by E

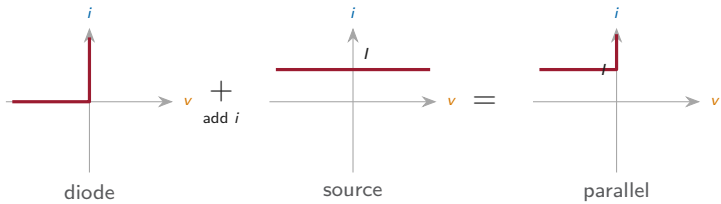
Case VI: a diode and a current source in parallel





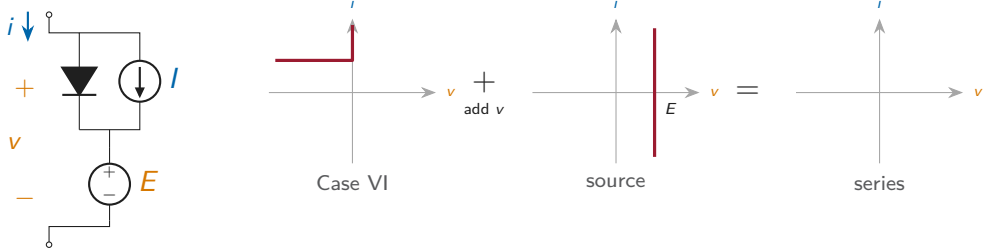
Case VI: a diode and a current source in parallel



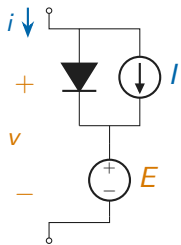


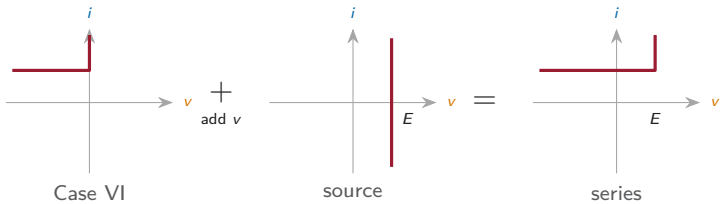
The diode curve shifted up by I

Case VII: sources and a diode



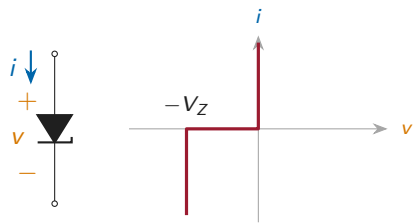
Case VII: sources and a diode





The source has $+$ at the top: shifted right by E

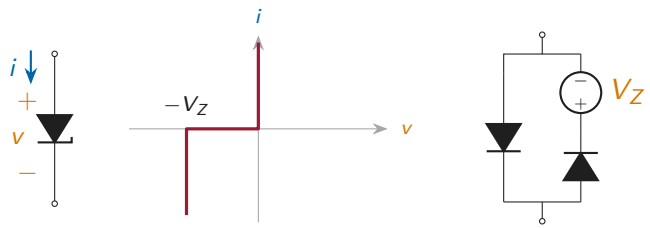
The ideal Zener diode



Forward: like a diode

Reverse: blocks until $v = -V_Z$

The ideal Zener diode

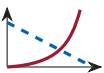


Forward: like a diode

Reverse: blocks until $v = -V_Z$

Two diodes and a battery V_Z

Roadmap



Load line

operating point ✓

none or many ✓



Substitution

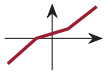
Thévenin first ✓



Ideal diode

on and off ✓

series, parallel ✓



Piecewise-linear circuits

4

i–*v* of a circuit

circuit from *i*–*v*

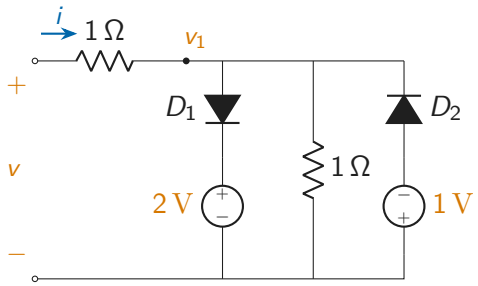


Time-varying

5

superposition

Example 15.4: two diodes



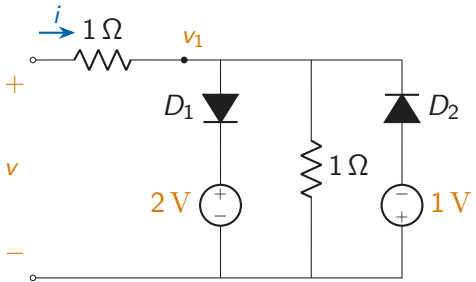
i as a function of v $= ?$

Two diodes: four combinations

One linear circuit for each

Keep each result where it is valid

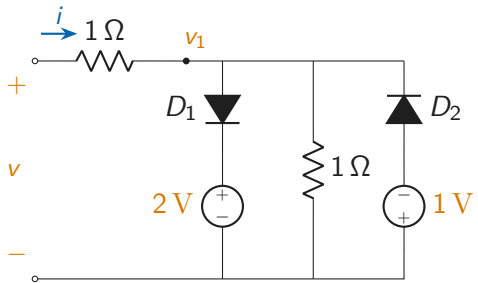
Example 15.4: both diodes OFF



$1\,\Omega$ and $1\,\Omega$ in series:

$v_1 = i, \quad v = 2\,i$

Example 15.4: both diodes OFF



1 Ω and 1 Ω in series:

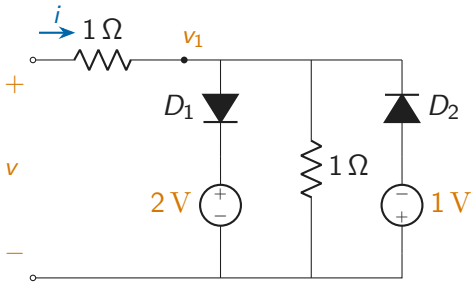
$v_1 = i, \quad v = 2\,i$

D_1 OFF: $v_{D_1} = v_1 - 2 < 0$

D_2 OFF: $v_{D_2} = -1 - v_1 < 0$

$-1 < v_1 < 2 \;\Leftrightarrow\; -2 < v < 4$

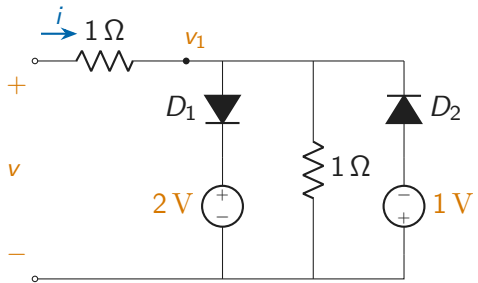
Example 15.4: D_1 ON, D_2 OFF



D_1 a short: $v_1 = 2$

$v = i + 2$

Example 15.4: D_1 ON, D_2 OFF



D_1 a short: $v_1 = 2$

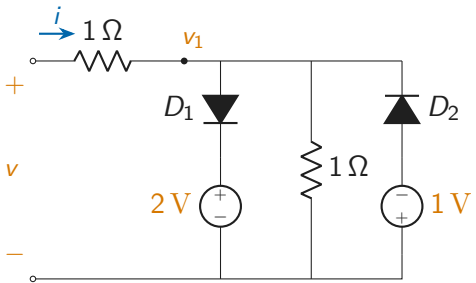
$v = i + 2$

middle $1\,\Omega$: $2\,\mathrm{A}$; D_1: $i_{D_1} = i - 2$

$i_{D_1} \geq 0 \;\Leftrightarrow\; v \geq 4$

D_2: $v_{D_2} = -1 - 2 < 0$ ✓

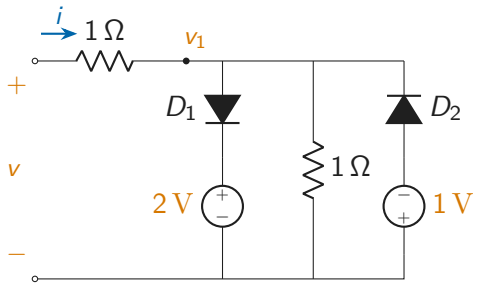
Example 15.4: D_2 ON, D_1 OFF



D_2 a short: $v_1 = -1$

$v = i - 1$

Example 15.4: D_2 ON, D_1 OFF



D_2 a short: $v_1 = -1$

$v = i - 1$

middle $1\,\Omega$: $-1\,\text{A}$; D_2: $i_{D_2} = -1 - i$

$i_{D_2} \geq 0 \;\Leftrightarrow\; v \leq -2$

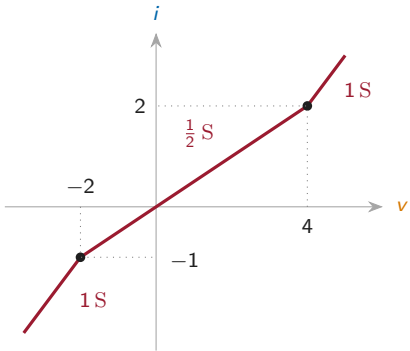
D_1: $v_{D_1} = -1 - 2 < 0$ ✓

Example 15.4: the result

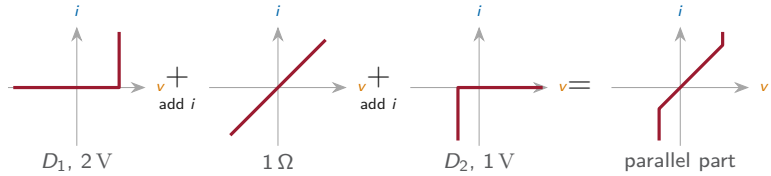
Both ON: $v_1 = 2$ and -1 (impossible)

$$i = \begin{cases} v - 2, & v \geq 4 \\ v/2, & -2 \leq v \leq 4 \\ v + 1, & v \leq -2 \end{cases}$$

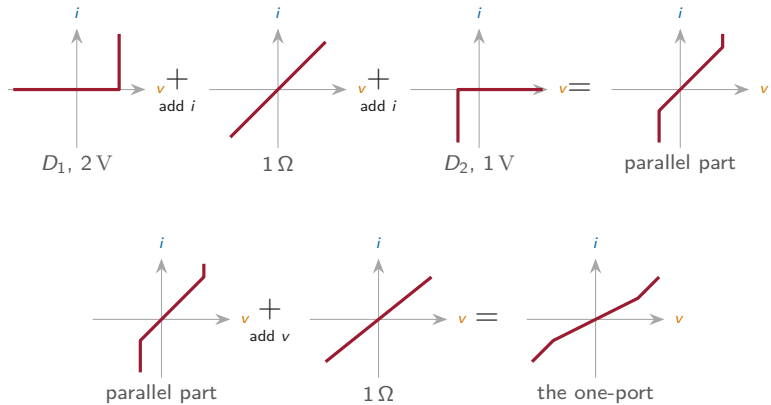
Two diodes: two breakpoints



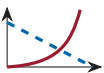
Example 15.4: graphically



Example 15.4: graphically



Roadmap



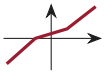
Load line



Substitution



Ideal diode



Piecewise-linear circuits



Time-varying

✓ ✓ ✓ 4 5

operating point ✓
none or many ✓

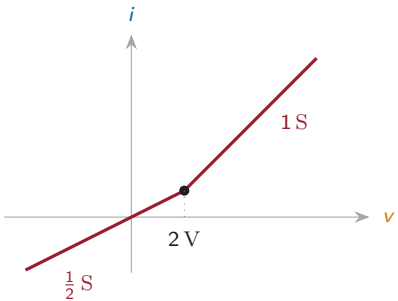
Thévenin first ✓

on and off ✓
series, parallel ✓

i–*v* of a circuit ✓
circuit from *i*–*v*

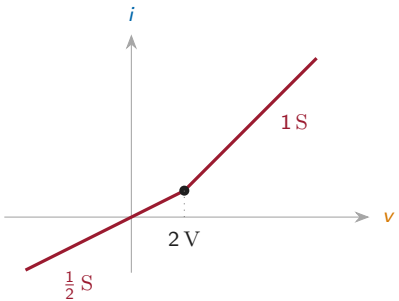
superposition

Example 15.5: a circuit from a characteristic



A circuit with this i–v characteristic $=$?

Example 15.5: a circuit from a characteristic

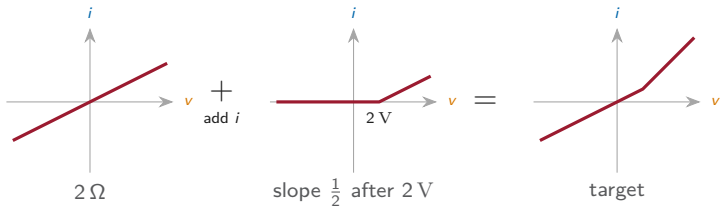


A circuit with this i–v characteristic $=$?

One breakpoint: one diode

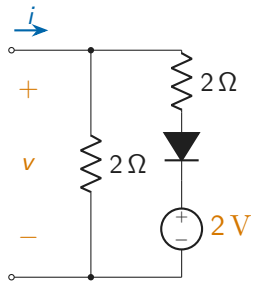
Below $2\,\mathrm{V}$: a line through the origin

Example 15.5: split the characteristic



- Slope $\frac{1}{2}\,\mathrm{S}$ everywhere: a $2\,\Omega$ resistor
- Extra slope $1 - \frac{1}{2} = \frac{1}{2}\,\mathrm{S}$ after $2\,\mathrm{V}$: Case I with $2\,\Omega$, shifted by $2\,\mathrm{V}$
- Currents add: the two in parallel

Example 15.5: the circuit



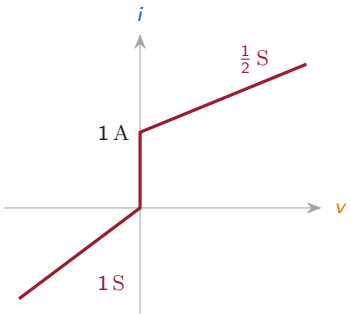
Check at $v = 4\,\mathrm{V}$:

$2\,\Omega$: $2\,\mathrm{A}$; branch: $\dfrac{4-2}{2} = 1\,\mathrm{A}$

$i = 3\,\mathrm{A}$ ✓

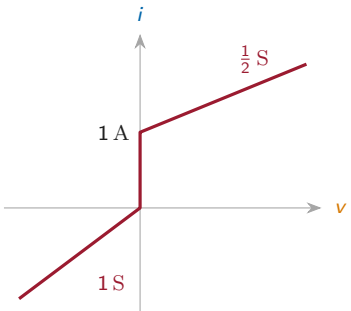
The order inside the branch does not matter: the circuit is not unique

Example 15.6: a circuit from a characteristic



A circuit with this i–v characteristic $=$?

Example 15.6: a circuit from a characteristic

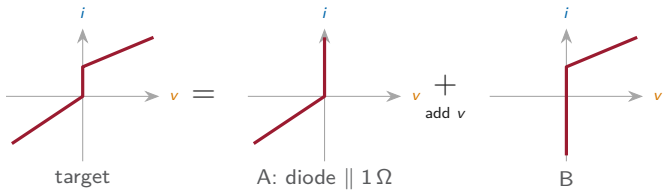


A circuit with this i–v characteristic = ?

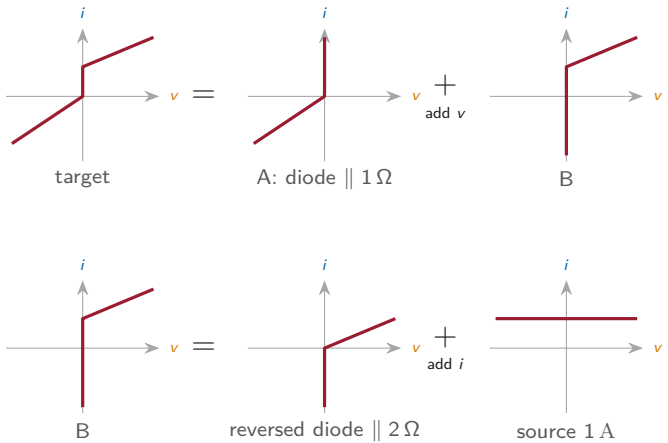
Two breakpoints: at least two diodes

Split into pieces we know

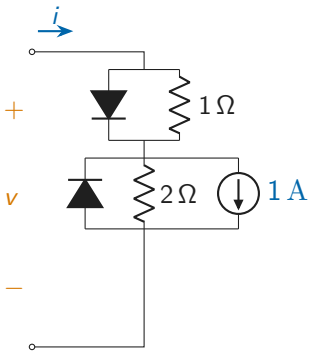
Example 15.6: split the characteristic



Example 15.6: split the characteristic



Example 15.6: the circuit



Top: A, the diode $\parallel 1\,\Omega$

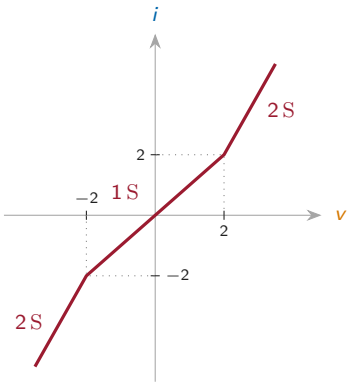
Bottom: B, reversed diode $\parallel 2\,\Omega \parallel 1\,\mathrm{A}$

A and B in series

Check at $v = 2\,\mathrm{V}$: $i = 1 + \frac{2}{2} = 2\,\mathrm{A}$ ✓

Another circuit may have the same characteristic: the circuit is not unique

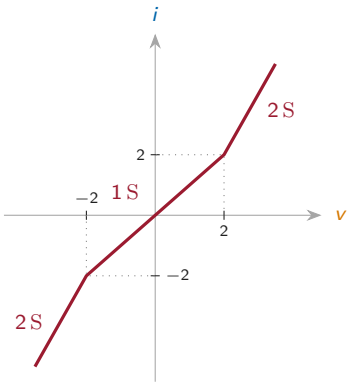
Example 15.7: a numerical characteristic



Slopes $2\,\mathrm{S}$, $1\,\mathrm{S}$, $2\,\mathrm{S}$

A circuit $=$?

Example 15.7: a numerical characteristic



Slopes $2\,\mathrm{S}$, $1\,\mathrm{S}$, $2\,\mathrm{S}$

A circuit $=$?

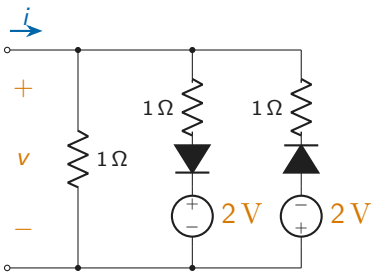
Split (parallel, add currents):

a $1\,\Omega$ resistor

$+$ a $1\,\mathrm{S}$ branch for $v > 2\,\mathrm{V}$

$+$ a $1\,\mathrm{S}$ branch for $v < -2\,\mathrm{V}$

Example 15.7: the circuit

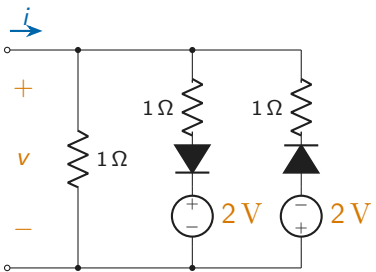


Branch 2: $v - 2$ for $v > 2\,\mathrm{V}$

Branch 3: turned around

$v + 2$ for $v < -2\,\mathrm{V}$

Example 15.7: the circuit



Branch 2: $v - 2$ for $v > 2\,\mathrm{V}$

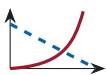
Branch 3: turned around

$v + 2$ for $v < -2\,\mathrm{V}$

Check at $v = 3\,\mathrm{V}$:

$3 + 1 + 0 = 4\,\mathrm{A}$ ✓

Roadmap



Load line

operating point ✓

none or many ✓



Substitution

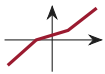
Thévenin first ✓



Ideal diode

on and off ✓

series, parallel ✓



Piecewise-linear circuits

i–v of a circuit ✓

circuit from i–v ✓

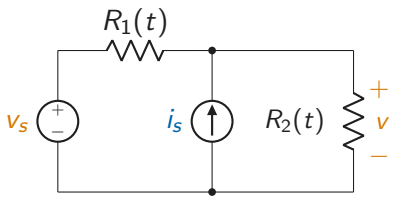


Time-varying

5

superposition

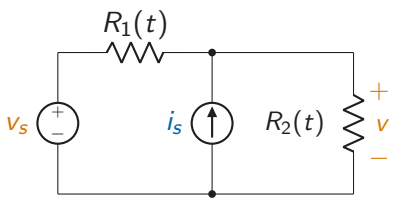
Example 15.8: a linear time-varying circuit



$R_1(t)$, $R_2(t)$: linear, time-varying

KCL: $\dfrac{v - v_s}{R_1} - i_s + \dfrac{v}{R_2} = 0$

Example 15.8: a linear time-varying circuit



$R_1(t)$, $R_2(t)$: linear, time-varying

KCL: $\dfrac{v - v_s}{R_1} - i_s + \dfrac{v}{R_2} = 0$

$$v = \underbrace{\frac{R_2}{R_1 + R_2}}_{k_1(t)} v_s + \underbrace{\frac{R_1 R_2}{R_1 + R_2}}_{k_2(t)} i_s$$

$v = \mathbf{k}(t)\,\mathbf{u}$: superposition still holds

Summary

In the circuit	What to do
one nonlinear resistor	load line $R\,i + v = E$ meets $i = g(v)$
a linear rest	Thévenin, operating point, substitution
ideal diodes	assume ON / OFF, solve, check
series, parallel	add voltages, add currents
a given characteristic	split it into known pieces
linear time-varying	$v = \mathbf{k}(t)\,\mathbf{u}$: superposition holds